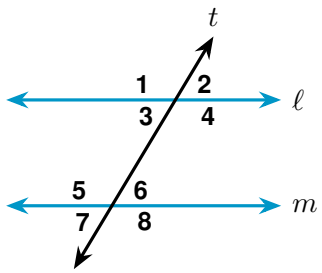
Proving Lines Parallel — Practice — Answer Key

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Is $\ell \parallel m$? Which Converse?

Watch Me — Worked Example



$m\angle 2 = 64°$ and $m\angle 6 = 64°$. $\angle 2$ and $\angle 6$ are corresponding and congruent.
Answer: yes, by the Corresponding Angles Converse.
If $m\angle 3 = 100°$ and $m\angle 5 = 70°$: same-side interior, $100 + 70 \neq 180$.
Not parallel.

Use the figure above. Is $\ell \parallel m$? If yes, name the converse.

1. $m\angle 3 = 75°$, $m\angle 6 = 75°$

yes; Alt. Interior Converse

2. $m\angle 1 = 120°$, $m\angle 5 = 120°$

yes; Corresponding Converse

3. $m\angle 4 = 80°$, $m\angle 6 = 100°$

yes; Same-Side Interior Converse

4. $m\angle 2 = 51°$, $m\angle 7 = 49°$

no

5. $m\angle 3 = 90°$, $m\angle 5 = 90°$

yes; Same-Side Interior Converse

6. $m\angle 1 = 132°$, $m\angle 8 = 132°$

yes; Alt. Exterior Converse

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Type 2 Find x So That $\ell \parallel m$

Watch Me — Worked Example

Alternate interior angles: $(6x - 4)^\circ$ and $(4x + 16)^\circ$. For $\ell \parallel m$ they must be congruent: $6x - 4 = 4x + 16 \Rightarrow x = 10$.

7. Corresponding angles: $(3x + 5)^\circ$ and $(5x - 21)^\circ$.

$x = 13$

8. Same-side interior angles: $(7x+2)^\circ$ and $(3x-2)^\circ$.

$x = 18$

9. Alternate exterior angles: $(8x-15)^\circ$ and $(6x+9)^\circ$.

$x = 12$

10. $m\angle 4 = (2x + 30)^\circ$, $m\angle 6 = (4x - 12)^\circ$.

$x = 27$

Type 3 Complete the Proof

Watch Me — Worked Example

Given $\angle 1 \cong \angle 5$. $\angle 1$ and $\angle 5$ are corresponding, so $\ell \parallel m$ (Corresponding Angles Converse).

11. Given: $\angle 2 \cong \angle 7$. Prove: $\ell \parallel m$.

Statements	Reasons
1. $\angle 2 \cong \angle 7$	Given
2. $\ell \parallel m$	Alternate Exterior Angles Converse

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12. HARD Lines $a \parallel c$ and $b \parallel c$. What can you say about a and b? Why?

$a \parallel b$: **parallel to the same line**

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